

Simulation of soil water dynamics in structured heavy soils with respect to root water uptake

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Abstract: In agricultural lands has the soil moisture uptake from the root system a significant effect on the water regime of the soil profile. In texturally heavy soils, where preferential pathways are present, infiltrated precipitation and irrigation water with diluted fertilizers quickly penetrate to a significant depth and often reach an under-root zone or even the ground-water level. Such a scenario is likely to happen during long summer periods without rain followed by heavy precipitation events, when a part of the water may flow through desiccated cracks.

Since 2001 the effects of drip irrigation and nitrogen fertilization of potatoes (*Solanum tuberosum* L., cultivar Agria) have been monitored within the frame of a research project at the experimental site Valecov (Czech Republic). Based upon the measured data an attempt has been made to simulate the water regime of the soil profile at a selected experimental plot, considering the impact of preferential flow and root water uptake. The dual-permeability simulation model S_1D_Dual (VOGEL et al., 2000) was used for the simulation. The soil hydraulic parameters were inversely determined using Levenberg-Marquardt method. Measured and simulated pressure heads were utilized in the optimization criterion. The scaling approach was applied to simplify the description of the spatial variability of the soil profile.

The results of simulations demonstrate that during particular rainfall events the water reaches significant depths of the soil profile via preferential pathways. The effect of the root zone is dominant during dry periods, when capillary water uptake from the layers below roots becomes important. This should be taken in account into the optimization of the drip irrigation and nitrogen fertilization schedule.

Key words: preferential flow, vadose zone, simulation, inverse modelling, root zone, dual-permeability approach

Introduction

In agricultural lands has the root water uptake a significant effect on the water dynamics within the soil profile during the vegetation season. Under randomly distributed precipitation, the potato crop usually exhibits some water stress. The amount of water for irrigation is usually limited, on the other hand high amounts of irrigation may cause a fast transport of fertilizers below the root zone. Thus an efficient irrigation regime is required to ensure a stable water supply of potatoes with a minimum negative effect on the environment. Several simulation models have been published which take the potato root growth and a variable water use into account (e.g. HAMER et al., 1994; SHAYKEWICH et al., 1998). However, none of these studies have considered the presence and consequences of preferential flow.

Soil macropores and parent rock heterogeneities appear to be among of the decisive factors of soil water regime on sites with a soil profile developed on acid crystalline rocks to which the experimental area be-

longs. Although these factors play also a crucial role in the irrigation management, they are rather difficult to observe and to include into the irrigation and fertilization schedule.

The aim of our study was to simulate the water regime of the soil profile for the vegetation season 2004 with respect to the mentioned soil heterogeneities and root water uptake, by means of a numerical model which allows to consider also the fast preferential flow.

Material and methods

The experiments with the aim to optimize drip irrigation and nitrogen fertilization of potatoes (*Solanum tuberosum* L., cultivar Agria) have been conducted at the research station Valecov (latitude 49°38'N, longitude 14°30'E, altitude 460–462 m a.s.l.), in the Bohemo-Moravian Highlands, Czech Republic, since 2001.

The soil, formed on the weathered paragneiss, exhibits a high variability of grain sizes. The soil has been classified as a deep and heavy pseudogley with a high ratio of clay particles (*Planosol*). The soil profile contains preferential

pathways, namely desiccated cracks, biopores and planar macropores on the boundaries of soil aggregates. The soil swells and shrinks, as was observed during the in-situ and laboratory dye-tracer experiments (JELÍNKOVÁ *et al.*, 2005).

Potatoes were grown on six plots under different conditions of drip irrigation, fertilization regime and crop rotation. The spacing between potato rows was 0.75 m and the seed tubers in rows were placed 0.35 m apart. A plot with no irrigation was chosen for simulations in order to express the potential crop water demand.

Potato plants produce a fibrous root system, which arises from initials along the underground portion of a stem. According to field observations the roots extend to depths of roughly 35 cm. Literature gives examples of the extraction of soil moisture from the depths down to 150 cm (CARR, 1988; STALHAM & ALLEN, 2001; FERREIRA & CARR, 2002). PARKER *et al.* (1989) reported water uptake from layers 0.3–0.4 m below the observed rooting depth on a sandy loam soil. At the experimental site the development of the root system is limited due to a heavy underlying layer 35 cm below the surface which is hardly penetrable for potato roots. Traces of mainly dead roots were observed up to depths of 60 cm in desiccated cracks filled with in-leaked small particles (DOLEŽAL *et al.*, 2004).

Climatic data were recorded at an automated meteorological station, which is located in the vicinity of the experimental field. The long-term average air temperature is 7.2°C and the average annual precipitation is 660 mm. The soil water pressure at depths 0.45 m and 0.75 m below the soil surface was measured with tensiometers, the readings were recorded hourly. The soil hydraulic properties of the soil matrix were measured on the core samples with the use of the plate apparatus and the outflow cells. The actual evapotranspiration was estimated according to FAO 56 directive (ALLEN *et al.*, 1998).

The one dimensional simulation of the soil regime was done by means of S_1D_DUAL simulation model (VOGEL *et al.*, 2000) based on the dual permeability approach, where the porous media is characterized by two continua system – the matrix domain and the preferential flow domain. Water flow in each domain is governed by Richards' equation, the contact between the domains is ensured by the transfer term (GERKE & VAN GENUCHTEN, 1993). To simplify the description of the spatial variability of the soil profile the scaling approach was applied (VOGEL *et al.*, 1991; DUŠEK *et al.*, 2006).

The parameters of the soil hydraulic functions of both domains and the values of scaling factors and of the transfer term were inversely determined in three separate consequent simulations, using Levenberg-Marquardt algorithm (DOHERTY *et al.*, 1995). The objective function was defined by means of measured and simulated pressure heads.

To acknowledge the time variable conditions in the soil profile caused by the root grow and the swelling of the soil, the simulated season 2004 was divided into three periods. The first 20 days (days 145–165), due to the low effect of the root zone and preferential pathways, were assumed as a reference period to determine the soil characteristics of the matrix domain. In other two periods (days 165–177 and 177–200, when the developing root zone and the preferential pathways due to desiccated cracks became important) the matrix domain characteristics were set unchanged and the saturated hydraulic conductivity of the preferential flow domain and the transfer term between the two domains were

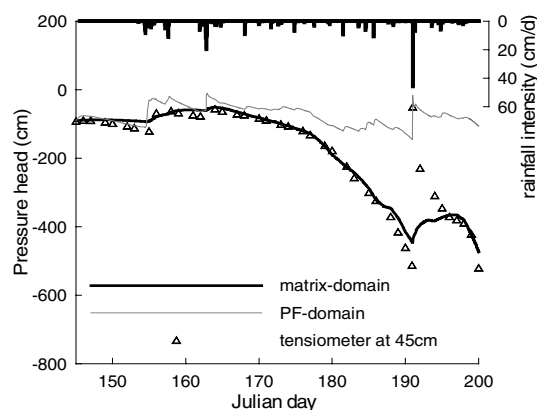


Fig. 1. Comparison of observed and simulated soil water pressure at a depth of 45 cm below soil surface.

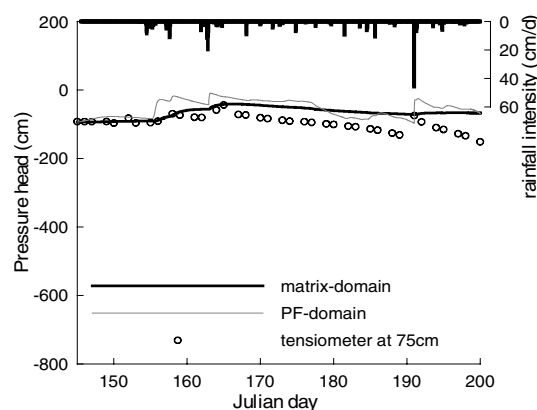


Fig. 2. Comparison of observed and simulated soil water pressure at a depth of 75 cm below soil surface.

inversely optimized. At these stages also the growth of the roots was involved in calculations.

Results and discussion

During the investigated period the total precipitation and evapotranspiration were approximately in balance (151 mm, resp. 158 mm). The soil water regime was affected by alternating of long rainless periods with short and heavy precipitation events. This phenomenon is apparent on Fig. 1, where at the 45 cm deep horizon the measured soil water pressure heads reached values close to –600 cm during the dry period and rapidly increased close to saturation immediately after a heavy rainfall.

The simulated pressure head development in the matrix domain fits the measured data rather well except of the event starting from day 190, when an extreme rainfall occurred. The difference between the measured and the simulated data can be explained by the water present in macropores. The installed tensiometer most likely intersects some of flooded preferential pathways. The measured values close to saturation correspond thus to the suctions in the preferential domain. Immediately after the event the matrix domain is lat-

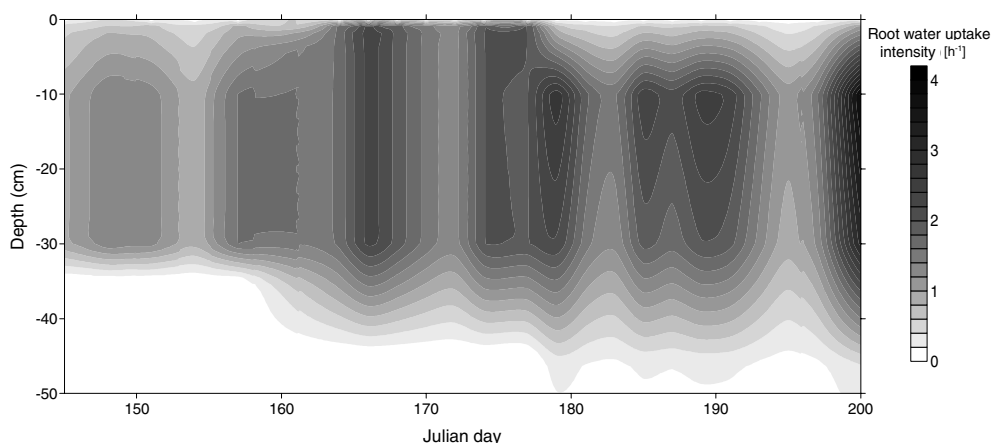


Fig. 3. Simulated root water uptake intensity along the soil profile during the growing period.

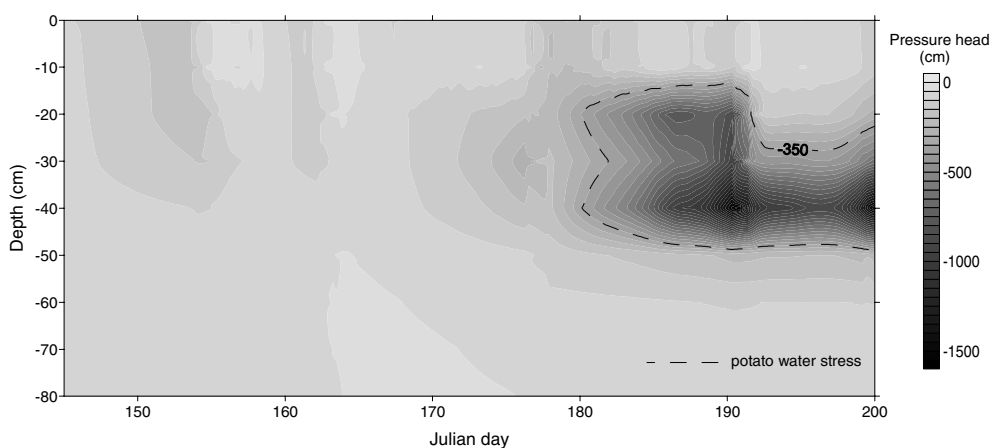


Fig. 4. Temporal changes of simulated pressure heads of the matrix domain along the soil profile within the simulated period.

erally saturated from macropores, which is well captured on the rapid rise of the simulated pressure head in the matrix domain, however the measured value corresponds to the simulated pressure head of the preferential flow domain (Fig. 1). As expected, the reactions of the preferential flow domain on the precipitation events are much more dynamic than in the matrix.

The simulated response to the root water uptake and the precipitation in the deeper layers is slow and rather stable. At a depth of 75 cm the pressure heads are almost constant. Figure 2 shows the slow decrease of measured pressure heads in the period without rain. The difference between the observed and simulated data might be caused by the neglecting of the root water uptake from the horizons deeper than 50 cm below the soil surface. The considered maximum rooting depth corresponds to field observations. The best fit of the measured and the simulated pressure heads is achieved in the first simulated period, when the effect of the root water uptake is not significant.

The map of intensities of the potato root water uptake within the simulated period is shown in Fig. 3. The uptake gradually became more significant with the growing roots and the rising evapotranspiration along

the season. The highest intensities of the soil moisture uptake by roots appeared at shallow depths up to 35 cm in the second half of the simulated growing season period.

The simulated pressure heads in the matrix domain along the soil profile within the simulated period are shown in Fig. 4. The dashed line represents the pressure head of -350 cm, which is assumed to be the value of the water stress. The area within the dashed border demonstrates conditions of the soil moisture shortage for the potato crop. Although the root water uptake takes place in the shallow horizon, due to the capillary rise the soil moisture is extracted from depths up to 30 cm below the potato roots.

Conclusions

The results of simulations (conducted for the vegetation season 2004 with no irrigation) proved an appreciable significance of preferential pathways on the soil water regime of the experimental plot under the study. In the dual domain approach the simulated pressure heads in shallow horizons exhibit fast reactions on particular precipitation events (Figs 1 and 4). If the dual

permeability approach has not been taken into account, the reactions of the model were much weaker. Simulations validate the need of irrigation in dry periods in order to keep the potato crop out of water stress.

In the case of fertilizing, water flows faster through the macropores and reaches deeper levels below the root zone. This fact must always be taken into account in designing of the drip irrigation and nitrogen fertilization regimes, especially for soils with a high ratio of clay particles. On the other hand, low rainfall intensities do not remarkably affect the water regime of the middle and lower parts of the soil profile and the effect of preferential pathways is during these periods negligible.

The effect of root water uptake on the water regime in the soil profile is significant. The model reacts sensitively on the changes of rooting depths and root distribution functions. In the simulations it was evident that even a very low root water uptake from deep layers strongly influences the water regime. The accuracy of the determination of the actual evapotranspiration is another important factor with an impact on the simulation results.

Acknowledgements

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